

Matrix multiplication

- ▶ can multiply $m \times p$ matrix A and $p \times n$ matrix B to get $C = AB$:

$$C_{ij} = \sum_{k=1}^p A_{ik} B_{kj} = A_{i1} B_{1j} + \cdots + A_{ip} B_{pj}$$

for $i = 1, \dots, m, j = 1, \dots, n$

- ▶ to get C_{ij} : move along i th row of A , j th column of B
- ▶ example:

$$\begin{bmatrix} -1.5 & 3 & 2 \\ & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3.5 & -4.5 \\ -1 & 1 \end{bmatrix}$$

Properties

- ▶ $(AB)C = A(BC)$, so both can be written ABC
- ▶ $A(B + C) = AB + AC$
- ▶ $(AB)^T = B^T A^T$
- ▶ $AI = A$ and $IA = A$
- ▶ $AB = BA$ *does not hold in general*

Block matrices

block matrices can be multiplied using the same formula, e.g.,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix}$$

(provided the products all make sense)

Column interpretation

- ▶ denote columns of B by b_i :

$$B = \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}$$

- ▶ then we have

$$\begin{aligned} AB &= A \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} \\ &= \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_n \end{bmatrix} \end{aligned}$$

- ▶ so AB is ‘batch’ multiply of A times columns of B

Inner product interpretation

- ▶ with a_i^T the rows of A , b_j the columns of B , we have

$$AB = \begin{bmatrix} a_1^T b_1 & a_1^T b_2 & \cdots & a_1^T b_n \\ a_2^T b_1 & a_2^T b_2 & \cdots & a_2^T b_n \\ \vdots & \vdots & & \vdots \\ a_m^T b_1 & a_m^T b_2 & \cdots & a_m^T b_n \end{bmatrix}$$

- ▶ so matrix product is all inner products of rows of A and columns of B , arranged in a matrix

Gram matrix

- ▶ let A be an $m \times n$ matrix with columns $a_1, \dots, a_n$
- ▶ the *Gram matrix* of A is

$$G = A^T A = \begin{bmatrix} a_1^T a_1 & a_1^T a_2 & \cdots & a_1^T a_n \\ a_2^T a_1 & a_2^T a_2 & \cdots & a_2^T a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n^T a_1 & a_n^T a_2 & \cdots & a_n^T a_n \end{bmatrix}$$

- ▶ Gram matrix gives all inner products of columns of A
- ▶ example: $G = A^T A = I$ means columns of A are orthonormal

Outline

Matrix multiplication

Composition of linear functions

Matrix powers

QR factorization

Composition of linear functions

- ▶ A is an $m \times p$ matrix, B is $p \times n$
- ▶ define $f : \mathbf{R}^p \rightarrow \mathbf{R}^m$ and $g : \mathbf{R}^n \rightarrow \mathbf{R}^p$ as

$$f(u) = Au, \quad g(v) = Bv$$

- ▶ f and g are linear functions
- ▶ *composition* of f and g is $h : \mathbf{R}^n \rightarrow \mathbf{R}^m$ with $h(x) = f(g(x))$
- ▶ we have

$$h(x) = f(g(x)) = A(Bx) = (AB)x$$

- ▶ composition of linear functions is linear
- ▶ associated matrix is product of matrices of the functions

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Matrix multiplication

Composition of linear functions

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Matrix powers

- ▶ for A square, A^2 means AA , and same for higher powers
- ▶ with convention $A^0 = I$ we have $A^k A^l = A^{k+l}$
- ▶ negative powers later; fractional powers in other courses